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**Dedicated To..**

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### NATURE-BIOINSPIRED MATERIALS AND MECHANISMS

#### **4.1 Echolocation:**

Echolocation is a biological or technological process that involves emitting sound waves and listening to the echoes that bounce back off of objects in the environment to determine their location, distance, and shape.

In biology, the use of echolocation by animals has been well known since the Ancient Greeks, for example, observed bats using echolocation to navigate and find food in the dark. The scientific study of echolocation in animals, however only began in the early 20<sup>th</sup> century, with the pioneering work of British naturalist Donald Graffin. Graffin's research showed that bats were using echolocation to navigate and hunt and helped modern study of biological echolocation.

In technology, the use of echolocation can be traced back to the early days of submarine warfare. During World War I, the British navy developed a primitive form of sonar (known then as "ASDIC") to detect submarines.

A comparison of biological and technological echolocation is given below:

#### **Biological Echolocation**

- ❖ Found in various animals such as bats, dolphins and some species of whales.
- ❖ Relies on the emission of sound waves, usually in the form of clicks or vocalizations.
- ❖ Animals emit sound wave, and listen for the echoes produced when the sound waves bounce off objects in their environment.
- ❖ By analyzing the echoes, animals can determine the location, distance, and even the shape of objects around them
- ❖ This ability is mainly used for navigation, hunting, and communication in the animal kingdom.
- ❖ Biological echolocation is a natural adaptation that has evolved over millions of years.

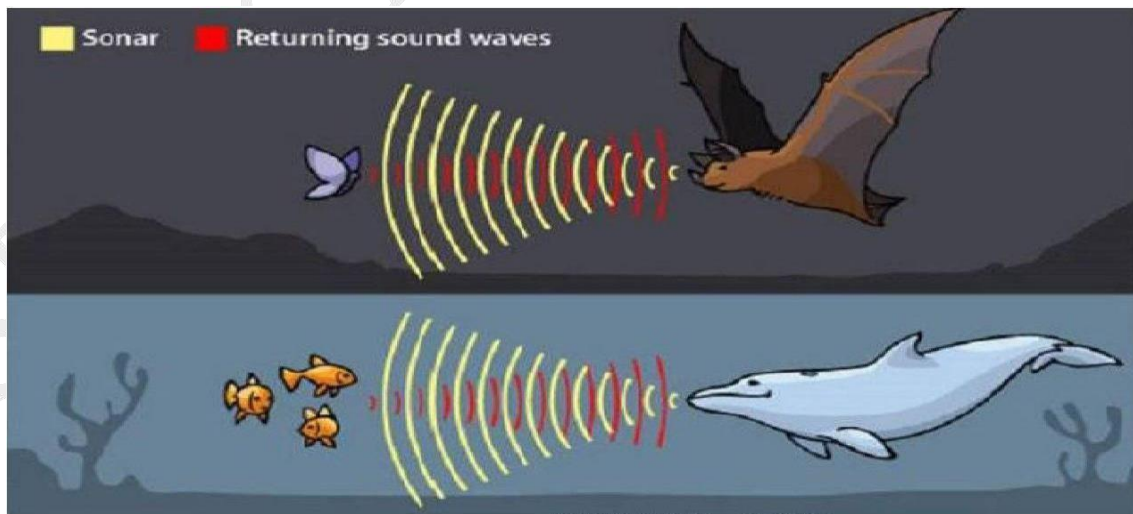
### **Technological Echolocation**

- ❖ Replicates the concept of biological echolocation using technological devices.
- ❖ Utilizes sound waves, typically generated by artificial sources such as sonar or ultrasonic sensors.
- ❖ These devices emit sound waves and analyze the echoes that bounce back from objects.
- ❖ The information from the echoes is processed and interpreted by the technology to generate useful data, such as distance, location, and object recognition.
- ❖ Technological echolocation has applications in various fields, including navigation, robotics, obstacle detection, and medical imaging.
- ❖ It is a human-engineered solution inspired by the natural abilities of animals

### **Principle of Ecolocation**

Both biological and technological echolocation rely on the same basic principles and have the same underlying purpose: to determine the location, distance, and shape of objects in the environment using sound waves and their echoes.

The principle of echolocation is based on the emission of sound waves and the interpretation of the echoes that bounce back from objects in the environment.



**Figure: Representing echolocation in bats and dolphins**

A concise explanation of the principle of echolocation is given in below

- ❖ **Sound Emission:** The echolocating organism, whether biological or technological, emits sound waves into its surroundings. In biological echolocation, this is typically achieved through vocalizations or clicks, while in technological echolocation, it is usually done using artificial sources such as sonar or ultrasonic sensors.
- ❖ **Propagation of Sound Waves:** The emitted sound waves travel through the environment, spreading out in all directions.
- ❖ **Object Interaction:** When the sound waves encounter objects in the environment, such as obstacles or prey, they interact with these objects. The interaction can involve reflection, scattering, or absorption of the sound waves.
- ❖ **Echo Reception:** Some of the sound waves that interact with objects bounce back or echo off them. These echoes carry information about the objects' distance, shape, composition, and other characteristics.
- ❖ **Sensory Reception:** The echolocating organism, whether biological or technological, has sensory receptors capable of detecting and processing the returning echoes. In biological echolocation, this is typically specialized organs or structures, such as bat ears or dolphin melons, while in technological echolocation, it is achieved through sensors and receivers.
- ❖ **Echo Interpretation:** The information contained in the echoes is analyzed and interpreted by the organism or technology. This interpretation involves extracting relevant features from the echoes and making sense of the spatial and temporal patterns present.
- ❖ **Perception and Response:** Based on the interpretation of the echoes, the organism or technology can perceive and understand the surrounding environment. This perception enables the organism to navigate, locate objects, detect obstacles, or perform other relevant tasks.

### Comparing the Sound Emission and Reception in Biological Ecosystem and Technological Ecosystem

In biological systems, sound emission and sensory reception organs are specialized adaptations that allow animals to engage in echolocation. Technological systems, on the other hand, employ devices designed to replicate and enhance these abilities.

Here's a concise comparison of sound emission and sensory reception organs/devices in biological and technological systems:

|                       | <b>Biological System</b>                                                                                                                                                                                                                                                                                                                                     | <b>Technological System</b>                                                                                                                                                                                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sound Emission</b> | <p>Biological organism such as bats, and cetaceans have specialized sound emission organs to produce sound for echolocation.</p> <p>Bats Emit sound using their larynx and modify the emitted sounds using structures like the nose leaf or mouth cavity.</p> <p>Dolphins and whales emit sounds through their blowholes, producing clicks or vocations.</p> | <p>Technological System rely on Artificial sound emission devices such as speakers or transducers to generate sound waves for echolocation.</p> <p>Ultrasonic sensors or sonar systems emits sound waves through these devices typically using piezoelectric elements transducers.</p> |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Sensory Reception</b></p> | <p>Biological organism posses specialized sensory reception organs that allow them to detect and interpret the returning echoes.</p> <p>Bats have Highly Sensitive ears designed to detect and analyze ultrasonic frequencies.</p> <p>Dolphins and some whales also receive echoes through their lower jaw. The jawbone conducts sound vibrations to the middle ear where they are converted into nerve impulse for interpretations by the brain.</p> | <p>Technologically System use Sensors and receivers to capture and process the returning echoes.</p> <p>Ultrasonic sensors are commonly employed which consist of a transducer that emits sound waves and receivers the echoes.</p> <p>Sonar system often incorporate hydrophones or other specialized underwater microphones to detect and interpret the echoes.</p> |
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### History of Technological Ecolocation

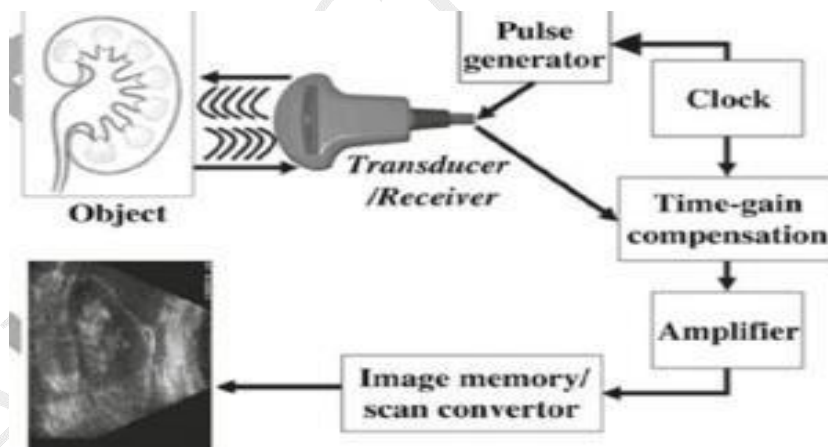
The history of technological echolocation can be traced back to the early development of sonar (sound navigation and ranging) technology. Here's a concise overview of the history of technological echolocation:

- ❖ **Early Sonar Development (late 19th century):** The foundations of technological echolocation were laid with the invention of the first practical underwater sound detection device called the hydrophone. Developed by Reginald Fessenden in the late 19th century, the hydrophone allowed for the detection of underwater sounds.
- ❖ **World War I (early 20th century):** During World War I, the need for detecting submarines led to significant advancements in sonar technology. Active sonar systems were developed, which involved the transmission of sound waves and the reception of echoes to detect submerged objects.
- ❖ **Further Advancements (mid-20<sup>th</sup> century):** The mid-20th century saw continued advancements in sonar technology, driven by military and Scientific research.

Sonar System were refined and improved for application such as submarine detection, underwater mapping and marine research.

- ❖ **Ultrasonic Application (mid-20<sup>th</sup> century):** In parallel with underwater sonar ultrasonic technology began to find application in fields such as medicine non-destructive testing, and industrial imaging. Ultrasonic sensors were developed for detecting and ranging objects based on the principle of echolocation.
- ❖ **Evolution of Echolocation Technologies (late 20<sup>th</sup> century present):** As Technology advanced more sophisticated echolocation system emerged Advancement in Signal processing, sensors and algorithm allowed for improved resolution, accuracy and interpretation of echoes. Echolocation technologies found application in various fields including robotics autonomous vehicles healthcare and environmental monitoring.

#### 4.1.1 Ultrasonography



**Figure: Representing working principle of ultrasonography**

- ❖ Ultrasonography is a medical imaging technique that uses high-frequency sound waves to produce images of the internal organs and tissues of the body.
- ❖ It is also known as ultrasound imaging or sonography.

- ❖ The ultrasound machine emits high-frequency sound waves (usually in the range of 2 to 18 MHz) that travel through the body and bounce back off of the internal organs and tissues.
- ❖ The returning echoes are captured by the ultrasound machine and used to create images of the internal structures.

Ultrasonography is a non-invasive, safe, and painless imaging method that can be used to visualize a wide range of structures within the body, including the organs of the abdomen, pelvis, and chest, as well as the uterus, fetus, and other soft tissues. It is commonly used in prenatal care to monitor the growth and development of the fetus and to diagnose any potential problems.

Ultrasonography has several advantages over other imaging methods, including its low cost, ease of use, and lack of ionizing radiation. It is also portable and can be used in a variety of settings, making it a valuable tool for medical professionals.

### Uses of Ultrasonography

Ultrasonography is a versatile imaging method that is used in a wide range of medical applications.

Some of the most common uses of ultrasonography include:

- ❖ **Obstetrics and gynecology:** Ultrasonography is commonly used to monitor the growth and development of a fetus during pregnancy, as well as to evaluate the reproductive organs and female pelvic organs for conditions such as ovarian cysts, fibroids, and endometrial cancer.
- ❖ **Abdominal imaging:** Ultrasonography is used to image the organs of the abdomen, such as the liver, gallbladder, pancreas, spleen, and kidneys. to diagnose conditions such as liver disease, gallstones, pancreatitis, and kidney stones.
- ❖ **Musculoskeletal imaging:** Ultrasonography is used to image the muscles, tendons, and ligaments to diagnose conditions such as muscle strains, tendonitis, and ligament sprains.

- ❖ **Vascular Imaging:** Ultrasonography is used to image the muscles, tenders and ligaments to diagnose condition such as blood clots, blockages, and aneurysms.
  - ❖ **Eye and neck imaging:** Ultrasonography is used to image the eyes and neck to diagnose condition such as cataracts, glaucoma and thyroid nodules.
- Emergency medicine: Ultrasonography is often used in emergency medicine to quickly and accurately diagnose condition such as appendicitis, pneumothorax, and fluid buildup in the abdomen or chest.

### Working Principle of Ultrasonography

The working principle of ultrasonography is based on the reflection of high-frequency sound waves.

- ❖ Transducers ultrasonography machine consists of a transducer that is used to emit and receive high-frequency sound waves. The transducer is placed in direct contact with the skin or inserted into the body through a gel.
- ❖ Emission of sound waves: The transducer emits high-frequency sound waves (usually in the range of 2 to 18 MHz) into the body. These sound waves travel through the body and encounter different tissues and organs, which have different acoustic properties.
- ❖ Reflection of sound waves: The sound waves encounter boundaries between different tissues and organs and bounce back, creating echoes. The strength of the echoes depends on the acoustic properties of the tissues and organs, such as density and stiffness.
- ❖ Reception of echoes: The transducer in the ultrasonography machine receives the echoes and sends the information to a computer, which processes the data to create images.
- ❖ Image formation: The computer uses the information from the echoes to create images of the internal organs and tissues of the body. The images are displayed on a screen, allowing the operator to see the structure and movement of the internal organs and tissues.

### Advantages of Ultrasonography

- **Non-invasive:** Ultrasonography does not involve any incisions or injections, making it a safe and convenient imaging method.
- **No ionizing radiation:** Ultrasonography does not use ionizing radiation making it a safer option for patients, especially pregnant women and children.
- **Real-time imaging:** Ultrasonography provides real-time images that can be used to monitor the movement and function of internal organs in real time.
- **Portable:** Ultrasonography machines are portable and making it a valuable tool for emergency and in
- **Cost-effective:** Ultrasonography is a cost-effective any special preparation or recovery time
- **Versatile:** Ultrasonography can be used to image a wide range of structures within the body, including the organs of the abdomen pelvis, and chest, as well as the uterus, fetus, and other soft tissues.

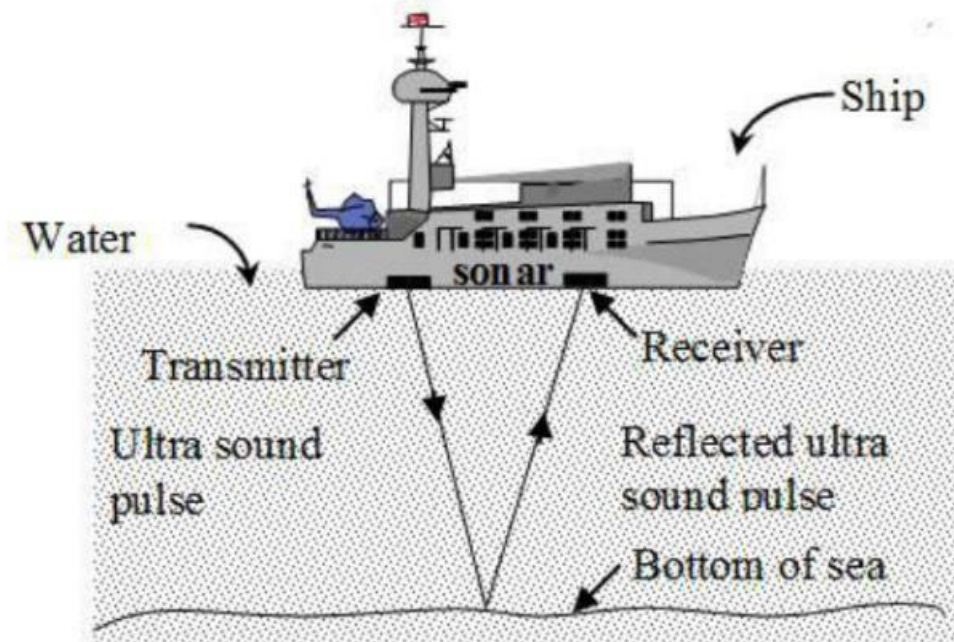
### Limitation of Ultrasonography

- ❖ **Limited depth:** Ultrasonography has limited depth and is not as effective at imaging deep structure or those obscured by bones or gas.
- ❖ **Operator dependence:** The Quality of the images produced by ultrasonography depends heavily on the skills and experience of the operator.
- ❖ **Limited Resolution:** Ultrasonography has limited resolution compared to other imaging methods making it less effective at visualizing small structures or detecting small changes in tissue.
- ❖ **Limitations in overweight patients:** Ultrasonography may have limited usefulness in overweight patients due to the difficulty in obtaining clear images through the layers of fat.

- ❖ **Limitation in detecting some types of cancer:** Ultrasonography may not be as effective at detecting certain types of cancer, such as pancreatic cancer, due to lack of characteristic signs on ultrasound images.

#### 4.1.2 Sonars

Sonar, which stands for Sound Navigation and Ranging, is a technology that uses sound waves to detect and locate underwater objects.



**Fig: Representing working principle of sonar**

#### Uses of Sonar

Sonars are commonly used for a variety of purposes, including;

- **Naval applications:** Sonars are used by naval vessels to detect and locate other ships, submarines, and underwater obstacles, allowing them to navigate safely and avoid potential collisions.

- **Fishery:** Sonars are used in the fishing Industry to locate schools of fish and determine the depth of the water, allowing fishermen to more efficiently target their catch.
- **Oceanography:** Sonars are used in oceanography to study the physical and biological properties of the ocean, including the structure of the ocean floor, the movement of currents, and the distribution of marine life.
- **Environmental monitoring** Sonars are used to monitor the health of marine ecosystems, track the migration patterns of whales and other marine mammals, and assess the impact of human activities on the ocean environment.
- Sonar technology works by emitting a series of sound pulses and listening for the echoes that bounce back from underwater objects. The time it takes for the echoes to return is used to calculate the distance to the objects, and the frequency and pattern of the echoes are used to determine their size and shape.

### Working Principle of Sonars

The working principle of sonar technology is based on the reflection of sound waves.

Here's how it works:

- **Transmitter:** A sonar system consists of a transmitter that produces and emits a series of sound pulses into the water. These sound pulses are typically in the form of high-frequency, low-power acoustic signals, known as "ping."
- **Propagation of sound waves:** The sound pulses propagate through the water, traveling to the target object and bouncing back as echoes. The speed of sound in water is slower than in air, and it depends on the temperature, pressure, and salinity of the water.
- **Receiver:** The sonar system also includes a receiver that listens for the returning echoes. The receiver is typically placed far away from the transmitter to minimize interference from the transmitted signals.

- **Calculation of range:** The time it takes for the echoes to return to the receiver is used to calculate the range to the target object. The range is simply the product of the speed of sound in water and the time it takes for the echoes to return.
- **Determination of target properties:** The Frequency and pattern of the echoes are used to determine the properties of the target object, such as its size, shape, and composition. For example, a large, solid object will produce a strong, low-frequency echo, while a small, porous object will produce a weaker, high-frequency echo.
- **Display of Results:** The Results of the sonar measurements are typically displayed on a screen or other output device, allowing the operator to visualize the target object and its location.

### Advantages of Sonar Technology

- **Versatility:** Sonar technology is versatile and can be used in a variety of applications, such as underwater navigation, mapping, and imaging, as well as for military and scientific purposes.
- **Cost-effective:** Compared to other underwater imaging technologies, sonar is relatively cost-effective and affordable.
- **Non-invasive:** Unlike other imaging technologies, such as diving and remote-operated vehicles, sonar does not physically disturb the underwater environment, making it an ideal choice for environmental monitoring and scientific research.
- **Real-time imaging:** Sonar provides real-time imaging allowing operators to quickly and easily assess the underwater environment.
- **High Resolution:** Modern sonar systems have high-resolution capabilities, allowing for detailed images of underwater objects and structures.

### Limitation of Sonar Technology

- **Limited visibility:** Sonar imaging is limited by the visibility of the water which can be affected by factors such as sediment, algae, and water temperature. This can make it difficult to obtain clear and accurate images.
- **Interference:** Sonar signals can be affected by interference from other underwater sources, such as ships, submarines, and natural underwater features, which can lead to false readings and reduced accuracy.
- **Short range:** Sonar signals have a limited range, which can make it difficult to image larger underwater structures or objects that are located far away from the sonar system.
- **Limited depth:** The depth to which sonar can effectively penetrate is limited, making it unsuitable for imaging objects or structures that are located at great depths.
- **Acoustic noise:** The use of sonar technology can also generate acoustic noise, which can disturb marine life and harm marine ecosystems. This is particularly a concern for high-power, military-grade sonar systems, which have the potential to cause serious harm to marine life.
- **Complex technology:** Sonar technology can be complex, requiring specialized skills and equipment to operate and maintain. This can limit its accessibility and increase the cost of implementation.
- **Inaccurate readings:** Sonar readings can be inaccurate due to factors such as reflection, refraction, and absorption of sound waves, which can result correct measurements and false readings.

## 4.2. Photosynthesis

Photosynthesis is the process by which plants, algae, and some bacteria convert light energy from the sun into chemical energy stored in organic molecules. This process is critical for life on Earth, as it provides the primary source of energy for all living organisms

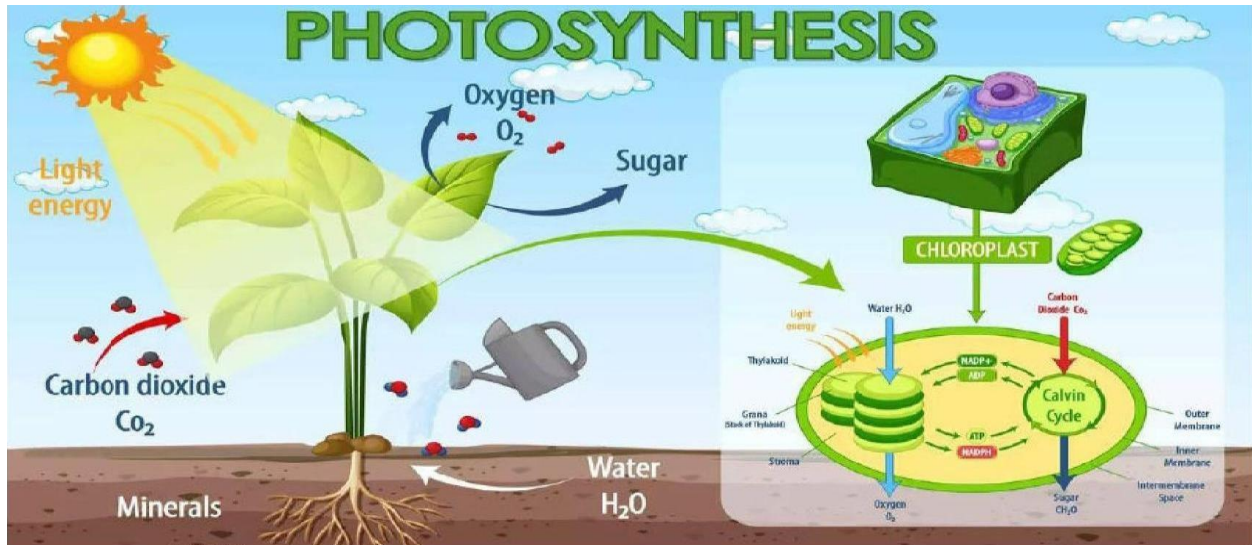


Figure: Representing photosynthesis

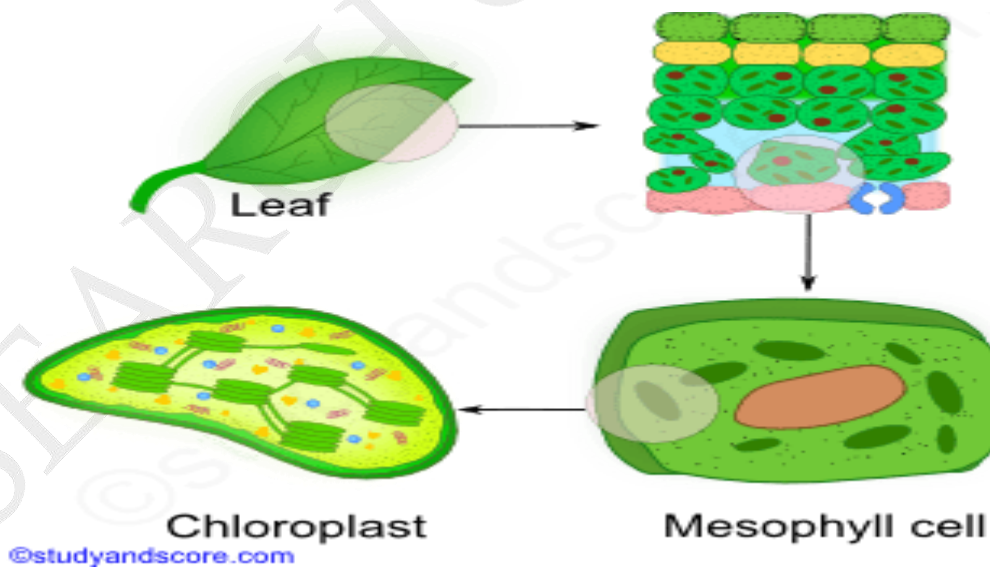


Fig: Indicating the mesophyll cell and chloroplast

### The Process of Photosynthesis in Plants and in Some Animals

The process of photosynthesis in plants and some animals differs in terms of the type of organisms involved and the specific details of the process. However, the basic principle of converting light energy into usable forms of energy is the same in both.

In plants, photosynthesis takes place in the chloroplasts of the cells located in the leaves. The process starts with the absorption of light energy by pigments such as chlorophyll, which then excites electrons. These excited electrons are used, to power the transfer of carbon dioxide into organic molecules, such as sugars and starches, through a series of chemical reactions. The end product of photosynthesis in plants is stored chemical energy in the form of organic compounds.

In some animals, such as algae, photosynthesis also takes place in chloroplasts. The process is essentially the same as in plants with the absorption of light energy and the conversion of carbon dioxide into organic molecules.

In contrast some animals, such as jellyfish, have a symbiotic relationship with photosynthetic organisms, such as algae. In this relationship, the animal provides a safe and stable environment for the photosynthetic organism, while the photosynthetic organism provides energy in the form of organic compounds produced through photosynthesis.

Light-dependent reactions and light-independent reactions (also known as the Calvin cycle) are two interconnected processes that occur in the chloroplasts of plants and algae during photosynthesis.

### **Light-Dependent Reactions:**

Light energy is absorbed by chlorophyll and other pigments in the thylakoid membranes of chloroplasts



Water molecules ( $H_2O$ ) are split through a process called photolysis, releasing electrons, protons ( $H^+$ ), and oxygen ( $O_2$ ).



The excited electrons from photolysis are captured by electron carriers, such as NADP<sup>+</sup> (Nicotinamide Adenine Dinucleotide Phosphate) and converted to NADPH (Nicotinamide Adenine Dinucleotide Phosphate).

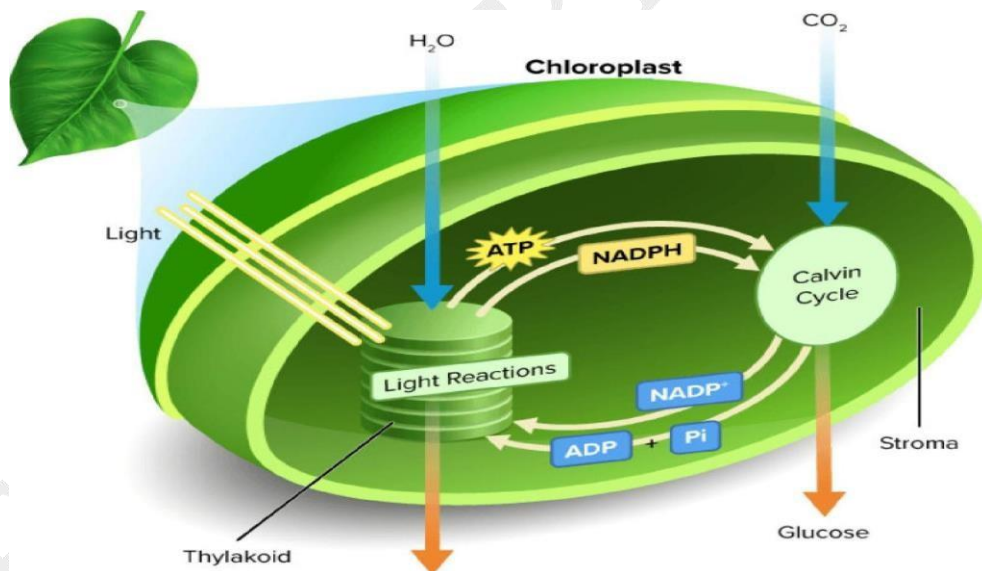


Adenosine diphosphate (ADP) combines with inorganic phosphate (Pi) to form adenosine triphosphate (ATP). This process is known as phosphorylation and is fundamental step in

cellular energy metabolism.



Oxygen molecules (O<sub>2</sub>) generated from the splitting of water are released as a byproduct into the atmosphere



**Fig: Representing chloroplast**

Carbon dioxide (CO<sub>2</sub>) from the atmosphere enters the stroma of the chloroplasts.



Carbon fixation occurs and forms a unstable six-carbon compound

↓

The resulting unstable six-carbon compound quickly breaks down into two molecules of 3-phosphoglycerate (PGA).

↓

ATP and NADPH, produced during the light-dependent reactions, provide energy and electrons for the reduction of PGA to glyceraldehyde-3-phosphate (G3P).

↓

Some G3P molecules are used to produce glucose and other organic molecules.

↓

The remaining G3P molecules are recycled and used in subsequent cycles of the Calvin cycle.

↓

Glucose and other carbohydrates produced during the Calvin cycle can be stored for later use or utilized in metabolic processes to meet the energy and growth needs of the plant.

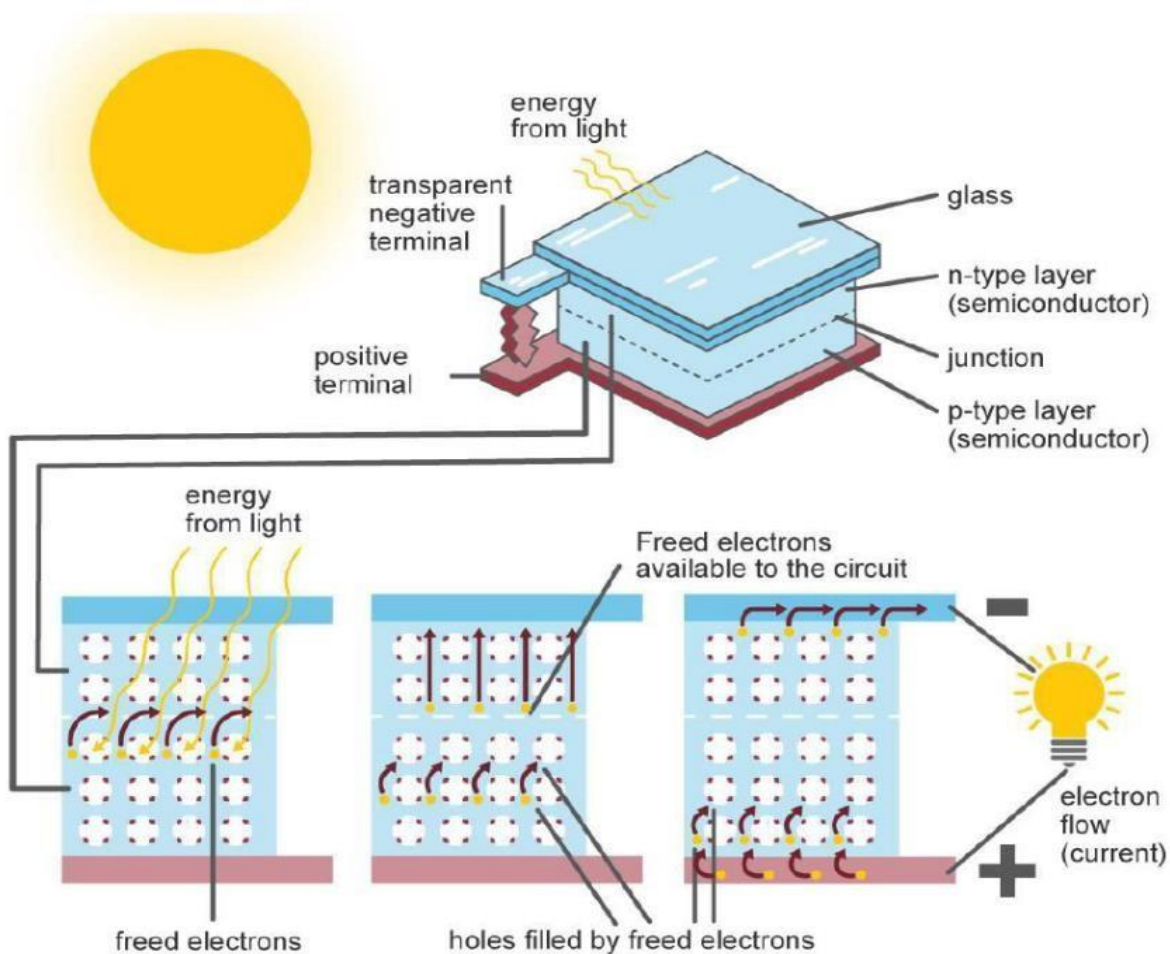
#### 4.2.1 Photovoltaic Cells

The connection between photosynthesis and photovoltaics lies in the conversion of light energy into usable forms of energy. In photosynthesis, light energy from the sun is converted into chemical energy stored in organic molecules, such as sugars and starches. In photovoltaics, light energy is converted into electrical energy.

Both photosynthesis and photovoltaics use the same basic principle of converting light energy into usable forms of energy, but the end products are different. In photosynthesis, the end product is stored chemical energy, while in photovoltaics, the end product is electrical energy.

However, the similarities between photosynthesis and photovoltaics go beyond just the conversion of light energy. Both processes also involve the use of specialized components and materials, such as chlorophyll in photosynthesis and silicon in photovoltaics, to absorb and convert light energy into usable forms of energy.

The development of photovoltaics has been heavily influenced by the natural process of photosynthesis, and many researchers have sought to mimic and improve upon the efficiency and effectiveness of photosynthesis in order to develop more advanced and efficient photovoltaic systems. The study of photosynthesis has thus played a significant role in the development of sustainable energy systems and continues to be an important area of research in the field of renewable energy.



**Fig: Representing working of a photovoltaic cell**

### New Technology Photovoltaic Cells

Photovoltaic cells, also known as solar cells, are devices that convert light energy from the sun into electrical energy. The technology behind photovoltaic cells has advanced significantly in recent years, leading to the development of new and improved photovoltaic cell designs and materials.

Some of the new technologies in photovoltaic cells include:

- **Perovskite solar cells:** Perovskite solar cells are a new type of photovoltaic cell that use a crystalline material made of perovskite to convert light energy into electrical energy. They are highly efficient and have the potential to be more affordable than traditional silicon-based photovoltaic cells.
- **Thin-film photovoltaic cells:** Thin-film photovoltaic cells are a type of photovoltaic cell that uses a thin layer of material, such as silicon or cadmium telluride, to convert light energy into electrical energy. They are lighter and more flexible than traditional silicon-based photovoltaic cells and are ideal for use in portable and flexible solar panels.
- **Concentrator photovoltaic cells:** Concentrator photovoltaic cells are a type of photovoltaic cell that uses a lens or mirror to concentrate sunlight onto a small area, increasing the amount of light energy that can be captured and converted into electrical energy.
- **Multi junction photovoltaic cells:** Multi junction photovoltaic cells are a type of photovoltaic cell that uses multiple layers of different materials, each optimized for different wavelengths of light, to convert light energy into electrical energy. They are highly efficient and ideal for use in concentrated solar power system.

These are just a few examples of the new technologies in photovoltaic cells. The field of photovoltaic is constantly evolving and there are many ongoing efforts to develop new and

improved photovoltaic cell designs and material that are more efficient, affordable, and environmentally friendly.

### 4.2.2. Bionic Leaf

A bionic leaf is a system that uses artificial photosynthesis to convert sunlight into usable forms of energy, such as hydrogen or other biofuels. The bionic leaf is designed to mimic the process of photosynthesis in plants, where light energy is used to split water molecules into hydrogen and oxygen, and the hydrogen can then be used as a source of energy.

The bionic leaf consists of a photovoltaic cell that captures sunlight and converts it into electrical energy, and a catalyst, such as a bacteria, that uses the electrical energy to split water molecules into hydrogen and oxygen. The hydrogen produced by the bionic leaf can then be stored and used as a source of energy for a variety of applications, such as powering vehicles or generating electricity.

The Bionic leaf has the potential to be a highly sustainable and environmentally friendly energy source, as it uses renewable resource such as sunlight and water to energy. Additionally, the bionic leaf can be used in remote locations where there is limited access to electricity and it can help to reduce our reliance on fossil fuels and mitigate the effects of climate change.

### Components of Bionic Leaf

A bionic leaf is a biohybrid system that mimics the natural process of photosynthesis to convert sunlight into chemical energy. It typically consists of several key components that work together to facilitate this conversion. Here are the main components of a bionic leaf:

- **Photosynthetic Organism:** The bionic leaf utilizes a photosynthetic organism, such as a cyanobacterium or a genetically modified plant, as the primary component. This organism contains chlorophyll or other light-absorbing pigments that capture solar energy and initiate the photosynthetic process.
- **Light Harvesting System:** The bionic leaf includes a light harvesting system, which can be artificial or natural, to efficiently capture sunlight. In some designs, light-absorbing dyes

or semiconductor materials are incorporated to enhance light absorption and conversion efficiency.

- **Catalysts:** The bionic leaf incorporates catalysts, such as enzymes (Examples: Hydrogenase, Nitrogenase, etc.) or synthetic catalysts (Example: Rubisco (Ribulose-1,5- biphosphate carboxylase/oxygenase)), to facilitate the chemical reactions involved in photosynthesis. These catalysts play a crucial role in splitting water molecules, generating electrons, and catalyzing the conversion of carbon dioxide into fuels or other chemical compounds.
- **Electron Transfer Pathway:** An electron transfer pathway is, an essential component of the bionic leaf system. It allows the generated electrons from water splitting to be efficiently transported to the catalysts involved in carbon dioxide reduction or other chemical reactions. This pathway ensures the flow of electrons necessary for fuel production or other desired chemical transformations.
- **Carbon Dioxide Source:** To sustain the photosynthetic process, a Bionic leaf requires a source of carbon dioxide. This can be obtained from various sources, including ambient air, industrial emissions, or concentrated carbon dioxide solutions.
- **Energy Storage or Conversion System:** The bionic leaf includes an energy storage or conversion system to capture and store the chemical energy produced during photosynthesis. This can involve the production of hydrogen gas, liquid fuels, or other energy-rich compounds that can be stored and used as needed.
- **Control and Monitoring System:** To optimize performance and ensure efficient operation, a bionic leaf typically incorporates a control and monitoring system. This system monitors various parameters such as light intensity, temperature, pH, and carbon dioxide levels, and allows for adjustments and optimization of the overall process.

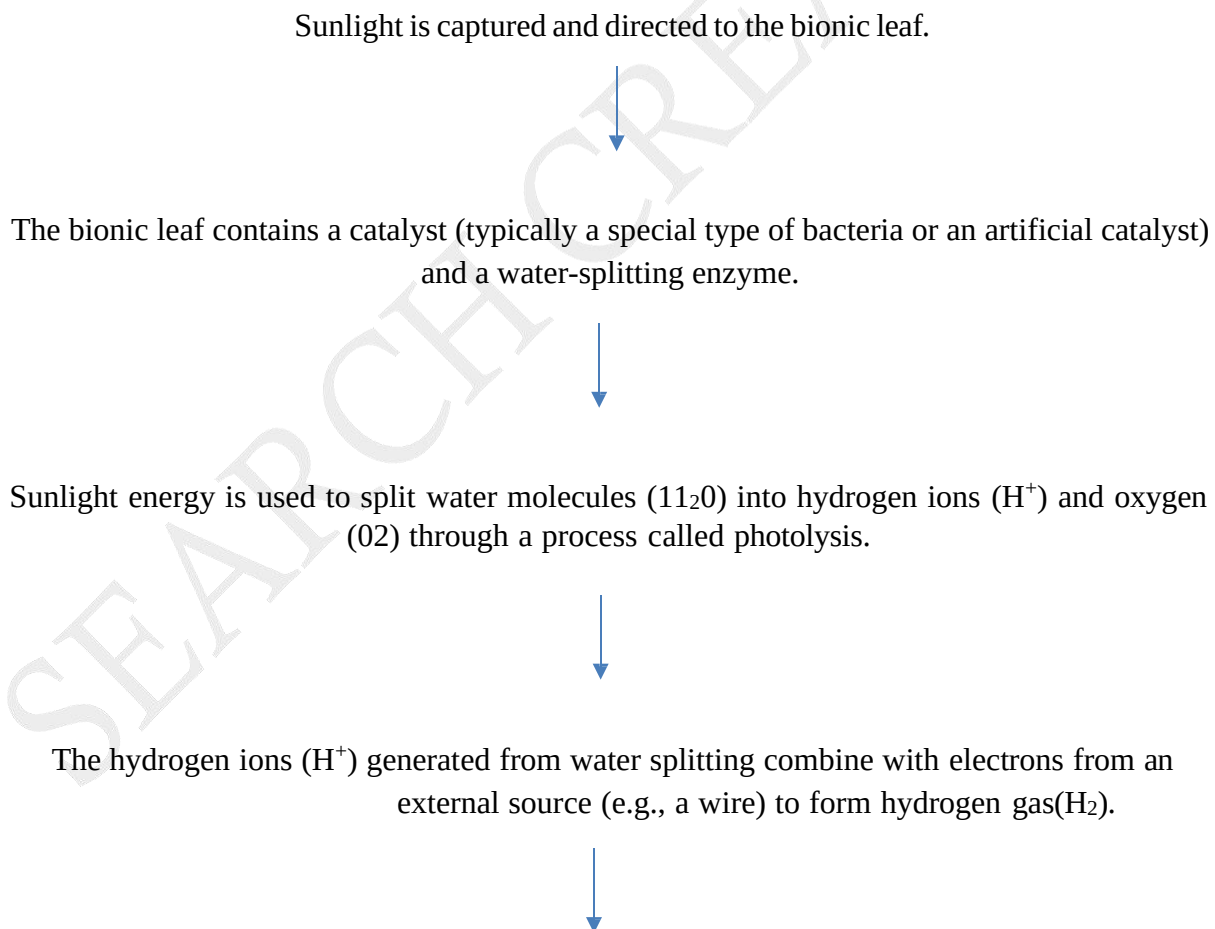
### Working Principle

The working principle of a bionic leaf is based on artificial photosynthesis, which aims to mimic the process of photosynthesis in plants. The bionic leaf typically consists of a photovoltaic cell that captures sunlight and converts it into electrical energy, and a catalyst, such as a bacterium that uses the electrical energy to split water molecules into hydrogen and oxygen.

The photovoltaic cell is used to convert sunlight into electrical energy, which is then passed to the catalyst. The catalyst, in turn, uses the electrical energy to power the process of water splitting, where water molecules are separated into hydrogen and oxygen. This process is facilitated by the presence of enzymes or other catalysts that act as a bridge between the electrical energy and the water splitting reaction.

The hydrogen produced by the bionic leaf can then be stored and used as a source of energy for a variety of applications, such as powering vehicles or generating electricity. Additionally, the oxygen produced by the bionic leaf can be released into the atmosphere, where it can help to mitigate the effects of climate change by reducing the levels of atmospheric carbon dioxide.

A flow chart of the working principle of bionic leaf is given below:



The generated hydrogen gas ( $H_2$ ) can be collected and stored for later use as a clean and renewable energy source.



The bionic leaf also absorbs carbon dioxide ( $CO_2$ ) from the air or a supplied source.



The Absorbed carbon dioxide ( $CO_2$ ) is converted into carbon-based compounds such as formic acid or methene, through a redirection reaction.



The carbon-based compounds can be used as a fuel or converted into other useful chemicals.



The Bionic leaf operates in a closed-loop system, where the produced oxygen ( $O_2$ ) during water splitting is reused by the catalyst in subsequent cycles.

### Applications of Bionic Leaf Technology

Here are some applications of bionic leaf technology:

- **Renew able Energy Production:** One of the primary applications of bionic leaf technology is in the production of renewable energy. Bionic leaf systems can harness solar energy and convert it into chemical energy in the form of hydrogen gas or other carbon-based fuels. These fuels can be used as clean energy sources for various applications, including transportation, electricity generation, and heating.
- **Carbon Dioxide Reduction:** Bionic leaf technology offers a promising solution for mitigating the rising levels of carbon dioxide in the atmosphere. By capturing and utilizing carbon dioxide as a feedstock, bionic leaf systems can potentially help reduce greenhouse gas emissions and combat climate change. This application holds significant potential for carbon capture and utilization (CCU) strategies.

- **Sustainable Chemical Production:** Bionic leaf systems can be utilized for sustainable chemical production. By utilizing carbon dioxide and renewable energy, these systems can produce a wide range of valuable chemicals, such as fertilizers, plastics, and pharmaceuticals. This application offers a more environmentally friendly and resource-efficient approach to chemical synthesis.
- **Agriculture and Food Production:** Bionic leaf technology can have applications in agriculture and food production. By utilizing sunlight and carbon dioxide, bionic leaf systems can generate oxygen and energy-rich compounds that can enhance plant growth and improve crop yields. This technology can potentially contribute to sustainable agriculture practices and help address global food security challenges.
- **Remote and Off-Grid Areas:** Bionic leaf systems can provide a decentralized and off-grid energy solution for remote or underdeveloped areas. By harnessing solar energy and producing clean fuels, these systems can offer sustainable power sources for communities without access to conventional energy infrastructure, enabling them to meet their energy needs and improve their quality of life.
- **Environmental Remediation:** Bionic leaf technology has the potential to aid in environmental remediation efforts. By utilizing the energy generated from sunlight, bionic leaf systems can power processes that remove pollutants or contaminants from air, water, or soil, contributing to the restoration and preservation of ecosystems.

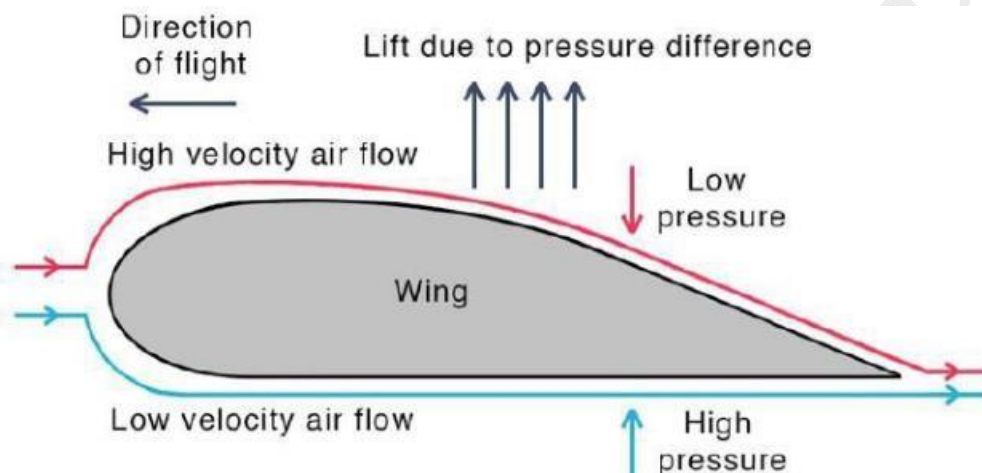
### 4.3. Bird Flying

Birds fly by flapping their wings and using their body weight and the movement of the air to stay aloft. They navigate using combination of visual cues, the Earth's magnetic field, and celestial navigation. Aircraft, on the other hand, use engines to generate thrust and lift from the wings to stay in the air navigate using a combination of instruments and systems, including GPS (Global Positioning System), which uses satellite signals to determine the aircraft's position and help it navigate. Although birds and aircraft both fly, their mechanisms and methods of navigation are quite different.

Birds flying influenced the invention of aircraft in that early aviation pioneers, such as the Wright brothers, observed and studied the flight of birds to develop their flying

machines. They noted how birds used their wings and body to achieve lift and control their flight, and used this knowledge to design and improve aircraft.

The development of GPS technology was not directly influenced by birds, but rather by the need for accurate and reliable navigation systems for various purposes, including aviation. GPS uses a network of satellites to provide location and time information, which is used by aircraft for navigation, communication, and safety purposes.



**Fig: Representing Bernoulli's Principle.**

The science behind the bird's flies using its wings and holding their body weight in air

The ability of birds to fly and support their body weight in the air is a result of various anatomical and physiological adaptations. Here's a Simplified Explanation of the science behind bird flight:

- **Wing Shape:** Birds have specialized wings with a unique shape that generates lift. The wings are curved on the upper surface and flatter on the bottom, creating a pressure difference known as Bernoulli's principle. This pressure difference generates lift, allowing birds to stay airborne.
- **Wing Muscles:** Birds have strong flight muscles attached to their wings, allowing them to flap their wings vigorously. The upstroke and downstroke motion of the wings generates thrust, propelling the bird forward through the air.

- **Hollow Bones:** Birds have lightweight bones that are hollow and filled with air sacs, reducing their overall weight. This makes it easier for them to stay aloft.
- **Feathers:** Feathers play a crucial role in flight. They provide both lift and control. The primary feathers at the tips of the wings help generate lift, while the tail feathers assist in maneuvering and stabilizing during flight.
- **Respiratory System:** Birds have a unique respiratory system that allows for efficient oxygen exchange. Air flows unidirectionally through their lungs, as well as through a system of air sacs located throughout their body. This constant supply of oxygen fuels their high metabolic demands during flight.
- **Efficient Circulatory System:** Birds have a highly efficient circulatory system that delivers oxygen-rich blood to their muscles and organs. Their heart rate increases during flight, ensuring a steady supply of oxygen to meet the demands of their active muscles.
- **Flight Control:** Birds have remarkable coordination and control over their flight. They can adjust the angle and shape of their wings, control their speed and direction, and perform intricate aerial maneuvers using their tail, wings, and body movements.

It's important to note that bird flight is a complex process influenced by several factors, including aerodynamics, muscle strength, metabolic efficiency, and specialized adaptations. The science behind bird flight continues to be an area of study and fascination for researchers and aviation engineers alike.

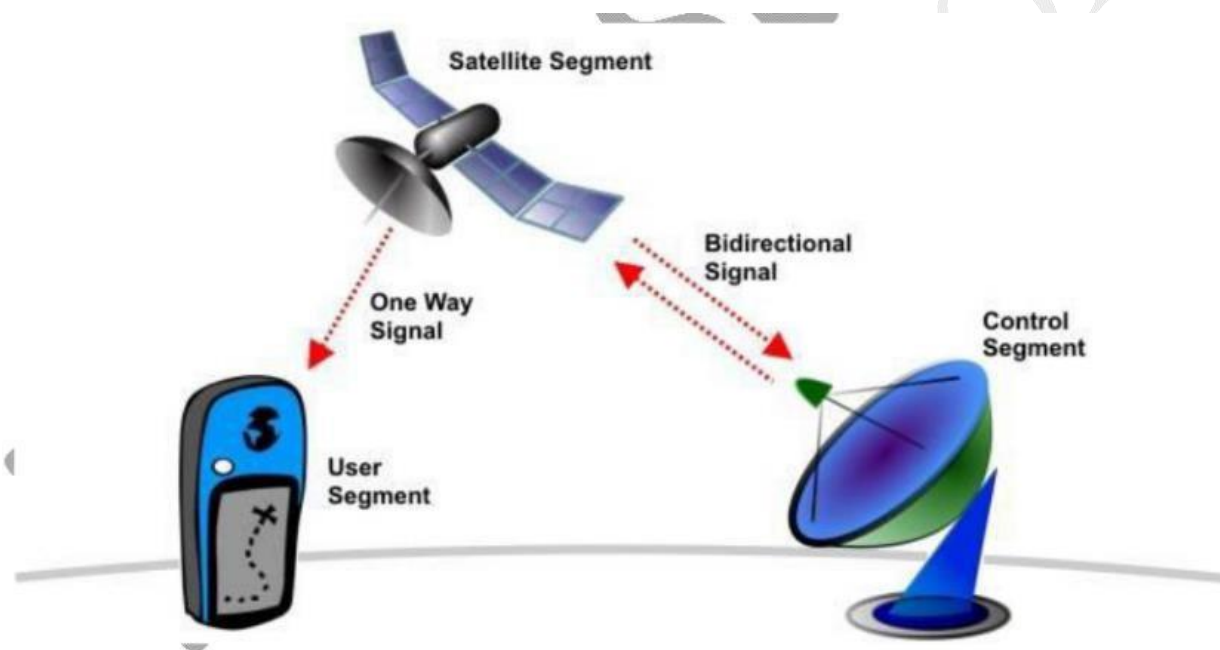
### 4.3.1 GPS Technology

GPS (Global Positioning System) is a technology that uses a network of satellites to provide location and time information to users. The technology works by measuring the time it takes for signals to travel from satellites to a receiver on the ground or in a vehicle, and using this information to calculate the user's position.

Here are some key components of GPS technology:

- **Satellites:** The GPS satellite network consists of 24-32 satellites orbiting the Earth. These satellites continuously broadcast signals containing information about their location, time, and status.

- **Receivers:** GPS receivers, which are typically integrated into devices such as smartphones, navigation system, and aircraft, receive signals from GPS satellite and use the information to calculate the user's position.
- **Control segments:** The control segments consist of ground-based monitoring stations that track the GPS satellite, check the accuracy of their signals and make adjustments as needed.
- **User segments:** The user segments consist of the GPS receivers used by individual and organization to obtain location and time information.



**Fig: Representing GPS**

GPS technology has a wide range of applications, including navigation, mapping, surveying, search and rescue, and military operations. The accuracy and reliability of GPS have improved over time, and the technology continues to evolve with new developments in satellite and receiver technology, as well as the integration of GPS with other technologies such as augmented reality and artificial intelligence.

### **Importance of GPS Technology in Aircrafts**



**Fig: Representing GPS technology in Aircrafts**

GPS technology is essential for aircraft navigation and guidance. Here's how it is used:

- **Positioning and Navigation:** GPS helps aircraft accurately determine their position and follow precise routes. Signals from satellites are received by GPS receivers onboard, allowing the system to calculate the aircraft's position.
- **Flight Planning:** GPS assists pilots and planners in creating optimal flight plans, considering waypoints, altitudes, and current information on navigation aids, weather, and airspace restrictions.
- **Approach and Landing:** GPS-based navigation systems provide precise guidance during approach and landing, even in low visibility. This enhances safety and reduces reliance on ground-based navigation aids.
- **Air Traffic Management:** GPS is integrated into air traffic management systems, improving airspace efficiency, reducing congestion, optimizing routing, and enhancing aircraft tracking and situational awareness for controllers.

- **Collision Avoidance:** GPS contributes to collision avoidance systems like TCAS and ADS-B. These systems use GPS data to track nearby aircraft, provide alerts, and ensure safe separation.
- **Flight Data Recording:** GPS data is often recorded by flight data recording systems, aiding post-flight analysis, accident investigation, and overall flight safety improvements. GPS technology has revolutionized aircraft navigation and has become an integral part of modern aviation. It provides accurate positioning, enhances safety, improves operational efficiency, and contributes to the overall advancement of the aviation industry.

**Comparing Birds and Aircrafts with GPS Technology for Navigation**

**Table:** Comparison between birds and aircraft with GPS technology for navigation

| Criteria  | Aircrafts                                                                                                                | Birds                                                                                                                                                                                                                                          |
|-----------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mechanism | GPS technology in aircraft relies on signals received from satellites to determine precise position, velocity, and time. | Birds use a combination of visual cues, magnetic fields, landmarks, and celestial navigation to navigate and orient themselves during flight.                                                                                                  |
| Accuracy  | GPS technology provides highly accurate position information with a margin of error typically within a few meters.       | Birds have remarkable navigational abilities but may not possess the same level of accuracy as GPS. However, birds can adjust their flight path based on real-time environmental cues, which allows for more dynamic and adaptable navigation. |

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|                     |                                                                                                                                  |                                                                                                                                                                                                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sensory Input       | GPS technology relies solely on receiving satellite signals.                                                                     | Birds integrate various sensory inputs for navigation. They can perceive and interpret visual cues, such as landmarks and the position of the sun or stars, and they may also have sensitivity to Earth's magnetic field, enabling them to navigate across vast distances.                                   |
| Adaptability        | GPS technology in aircraft provides consistent and reliable navigation regardless of the environmental condition or time of day. | Birds, on the <b>other</b> hand, demonstrate remarkable adaptability in their navigation abilities. They can adjust their flight paths based on changing weather conditions, wind patterns, and other factors, which allows for efficient long-distance migration and navigation through complex landscapes. |
| Evolutionary Aspect | GPS technology is a human-made innovation designed to enhance navigation and safety in aircraft.                                 | Birds, however, have evolved over millions of years, developing specialized neural and physiological adaptations that enable them to navigate and fly efficiently in diverse habitats.                                                                                                                       |

### 4.3.2 Aircraft Technology

Aircraft technology has advanced significantly since the first powered flight by the Wright brothers in 1903. Here are some key components of modern aircraft technology:

- ❖ **Aerodynamics:** Modern aircraft are designed to be more aerodynamic, with wing shapes optimized for lift and efficiency. Advanced materials and manufacturing techniques have also been developed to reduce weight and improve durability.
- ❖ **Jet engines:** Jet engines, which use the principles of Newton's third law of motion to produce thrust, have replaced propeller engines in most modern aircraft. These engines are more powerful, fuel-efficient, and reliable.
- ❖ **Avionics:** Avionics, or aviation electronics, have advanced significantly with the development of digital technology. Flight instruments, navigation systems, and communication systems have become more precise, reliable, and sophisticated.

- ❖ **Safety systems:** Aircraft safety systems have been developed to reduce the risk of accidents and improve passenger safety. These include systems for collision avoidance, weather detection, and emergency response.
- ❖ **Automation:** Aircraft automation has increased significantly in recent years, with the development of advanced autopilot systems and computerized flight control systems. This technology has made flying safer and more efficient, but has also raised concerns about pilot training and the potential for overreliance on automation.

### Bio Mimicking Birds Fly for Aircraft Technology

Biomimicry, or the practice of using designs and processes found in nature to solve human problems, has led to the development of various technologies inspired by birds' flight. Some examples include:

- ❖ **Wing design:** The shape of bird wings has inspired the design of aircraft wings, which have evolved to be more aerodynamic and fuel-efficient as a result. The study of bird flight has also led to the development of winglets, small structures at the tip of wings that reduce drag and increase lift.

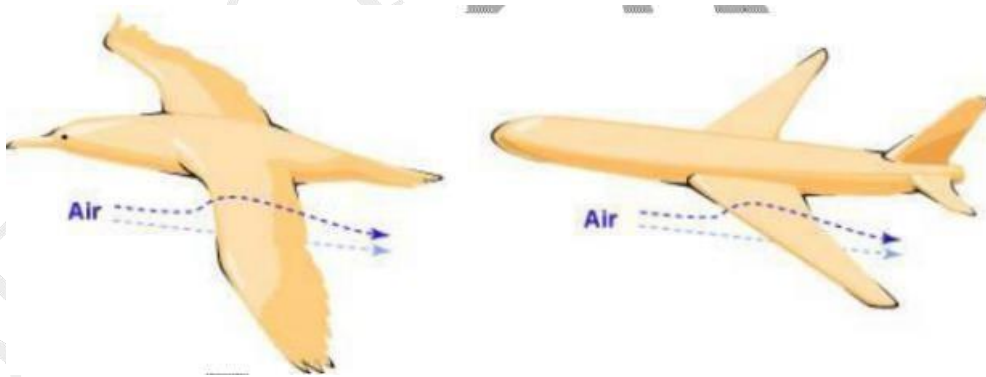


Fig: Comparing the wings design of bird and aircraft

- ❖ **Flapping-Wing drones:** Researches have developed drones that use flapping wings to fly, mimicking the way birds and insects fly. These drones can be used for various application,

such as monitoring crops and wildlife, inspecting building and infrastructure, and search and rescue operations.



**Fig: Image of a flapping-wing drone.**

- ❖ **Soaring algorithms:** Soaring refers to the flight technique used by birds and certain aircraft to stay aloft and travel long distances with minimal energy expenditure. It involves utilizing rising air currents, such as thermals, ridge lift, wind shear, or atmospheric waves, to gain altitude and maintain flight. Birds use thermals, or columns of rising warm air, to gain altitude and soar. Researchers have developed algorithms inspired by bird flight to help gliders and other aircraft use thermals more efficiently, leading to longer and more sustainable flights.
- ❖ **Landing gear:** The legs and feet of birds have inspired the design of landing gear for aircraft, with shock-absorbing and retractable structures that help absorb impact upon landing.

### **The Future of Transportation through the Air**

The future of transportation through the air hold exciting possibilities with the emergence of new technologies and concepts. Here are potential modes of air transportation that could shape the future:

1. **Electric Vertical Takeoff and Landing (eVIOL) Aircraft:** these are electric-powered aircraft that can take off and land vertically, similar to helicopters. They are being designed for urban air mobility and short-distance transportation, offering a more efficient and environmentally friendly alternative to traditional helicopters.

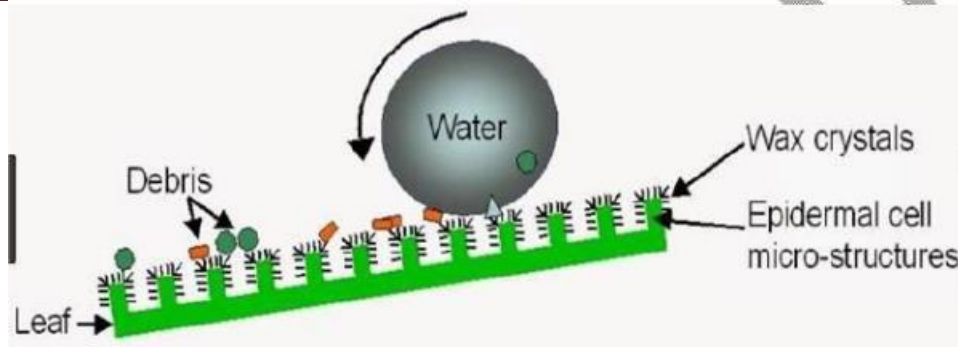
2. **Autonomous Flying Vehicles:** Autonomous drones a. 'flying taxis' are being developed for various applications, including transportation of people and goods. These vehicles would operate without a pilot and rely on advanced sensors, artificial intelligence, and automation to navigate safely.
3. **High-Speed Air Travel:** Supersonic and hypersonic aircraft are being explored to revolutionize long-distance travel. These aircraft would travel at extremely high speeds, significantly reducing travel times and opening up new possibilities for global connectivity.
4. **Personal Air Vehicles (p Vs):** PAVs are compact flying vehicles designed for individual use. They could potentially serve as a convenient mode of transportation for short- distance travel within cities, similar to personal cars but in the air.
5. **Hyperloop Transportation:** While not strictly an air-based mode of transportation, the hyperloop concept involves high-speed capsules traveling through low-pressure tubes, o g near-supersonic speeds. This mode of transportation could connect distant cities and regions in a fast, energy-efficient manner.

### 4.4 Lotus Leaf Effect

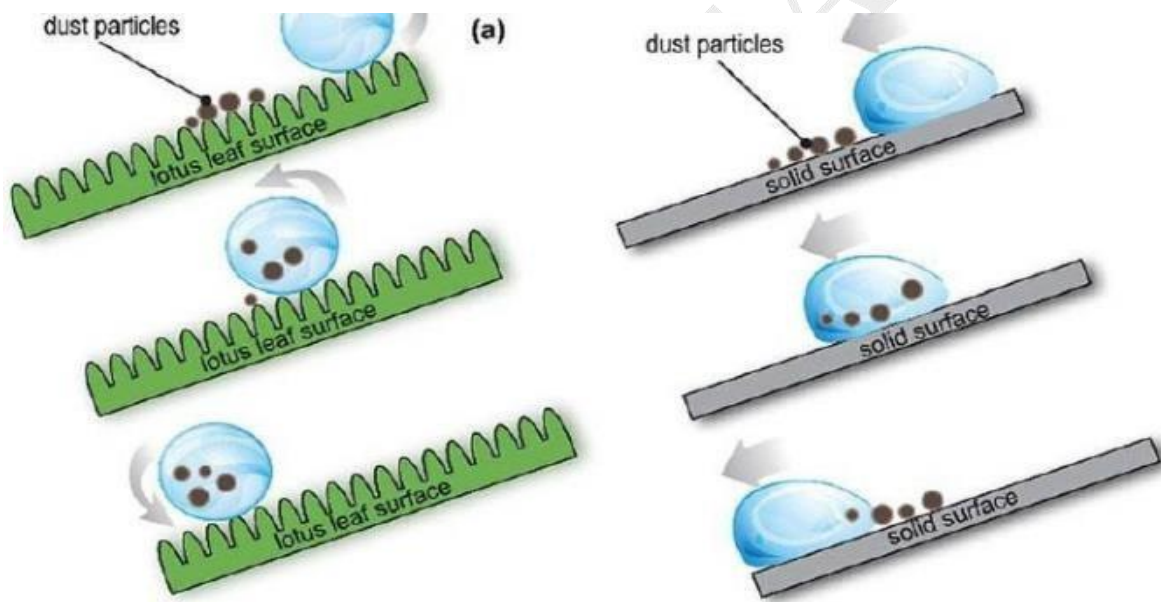
#### **Introduction**

The lotus leaf effect, also known as the "lotus effect," refers to the ability of lotus leaves to repel water and self-clean through their unique surface structure. This effect has inspired the development of super hydrophobic and self-cleaning surfaces, which have a wide range of applications in various industries.

The lotus leaf surface has a microscale and nano scale structure that consists of numerous small bumps and wax-coated hairs. This structure creates a high contact angle between the water droplets and the surface, causing the droplets to roll off and carry away Iv dirt or debris. This self-cleaning property is due to the lotus leaf's ability to repel water and resist adhesion.



**Fig: Representing the surface of lotus leaf**



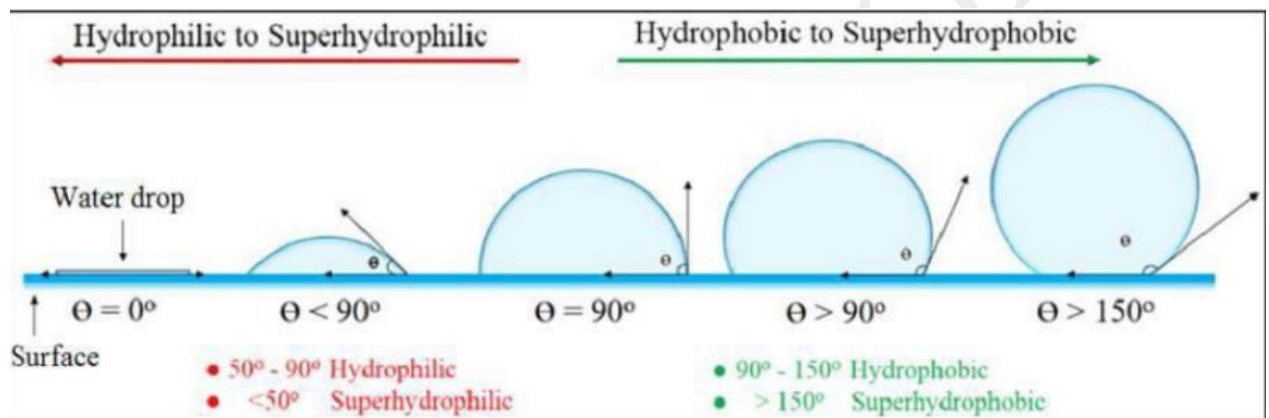
**Fig: Representing the behavior of water drops on slanted surface of  
a) A lotus leaf surface, and b) any other solid surface.**

Super hydrophobic and self-cleaning surfaces have applications in industries such as aerospace, automotive, building materials, and medical devices. For example, self-cleaning coatings can be used on the exterior of buildings to reduce the need for cleaning and maintenance, while super hydrophobic coatings can be used to prevent icing on aircraft wings.

#### 4.4.1. Super Hydrophobic Effect

##### The Principle of Super hydrophobic Surfaces

The super hydrophobic effect refers to the ability of certain surfaces to repel water and resist wetting. Super hydrophobic surfaces are characterized by a high contact angle between water droplets and the surface, typically over 150 degrees, and a low contact angle hysteresis, meaning that the droplets roll off the surface with ease.



**Fig: Representing super hydrophobic and super hydrophilic effects.**

##### Materials and Examples

Super hydrophobic surfaces are created by modifying the surface chemistry and structure of materials to achieve extremely high-water repellency. Several materials and coating techniques are used to prepare super hydrophobic surfaces. Here are some commonly used materials and examples:

- ❖ **Fluoropolymers:** Fluoropolymer-based coatings are widely used for super hydrophobic surfaces due to their low surface energy and water-repellent properties. Examples include polytetrafluoroethylene (PTFE) and fluorinated ethylene propylene (FEP) coatings.
- ❖ **Silica-based Nanoparticles:** Silica nanoparticles can be functionalized and applied to super hydrophobicity. These nanoparticles create a rough surface structure that trap air pockets, preventing water from wetting the surface. Additionally, the surface

can be modified with hydrophobic molecules. Examples include silica nanoparticles coated with hydrophobic agents like alkylsilanes.

- ❖ **Carbon-based Materials:** Carbon nanotubes (CNTs), graphene, and carbon nanofibers are used to create super hydrophobic surfaces. These materials can be aligned or randomly distributed to form a rough surface with hydrophobic properties. The combination of their unique structures and hydrophobic coatings contributes to water repellency.
- ❖ **Metal-based Materials:** Various metals and metal oxides can be used to create super hydrophobic surfaces. One approach involves creating micro/nanostructured surfaces using etching techniques, such as chemical etching or electrochemical etching, on metals like aluminum, copper, or stainless steel. These structures, combined with appropriate surface treatments, enhance water repellency.
- ❖ **Polymer-based Materials:** Some polymers, when processed and structured appropriately, can exhibit super hydrophobic properties. For example, polydimethylsiloxane (PDMS) can be modified and structured to create rough surfaces with low surface energy, resulting in super hydrophobic behavior.
- ❖ **Natural Materials:** Certain natural materials, such as lotus leaves and butterfly wings, have inherently super hydrophobic properties. Researchers have studied the surface structures and chemical composition of these natural surfaces to replicate them artificially. Mimicking the hierarchical structures and utilizing hydrophobic coatings can create super hydrophobic surfaces.
- ❖ **Hybrid Materials:** Combinations of different materials are often used to create super hydrophobic surfaces. For instance, hybrid coatings can be formed by combining nanoparticles, polymers, and other materials to achieve synergistic effects and optimize super hydrophobic properties.

### Techniques used to prepare super hydrophobic surfaces

To prepare super hydrophobic surfaces, various techniques are employed to modify the surface structure and chemistry of materials. These techniques aim to create roughness and reduce surface energy, leading to high water repellency. Here are some commonly used techniques:

- **Chemical Vapor Deposition (CVD):** CVD involves the deposition of thin films onto a substrate through chemical reactions in the vapor phase. By using appropriate precursors, surface coatings with low surface energy can be achieved, resulting in super hydrophobicity.
- **Sol-Gel Method:** The sol-gel process involves the synthesis of inorganic materials from a solution (sol) that undergoes a gelation process to form a solid network. By controlling the composition and structure of the sol-gel materials, super hydrophobic coatings can be created on various substrates.
- **Electrochemical Methods:** Electrochemical techniques like anodization and electroplating can be employed to create super hydrophobic surfaces. Anodization involves the controlled oxidation of metals, such as aluminum, to form a porous oxide layer with a rough surface. Electroplating can be used to deposit metals or alloys with desired surface properties.
- **Plasma Treatment:** Plasma treatment involves exposing the material surface to low-pressure plasma, which can modify the surface chemistry and morphology. Plasma etching, deposition, or functionalization techniques can be used to create superhydrophobic surfaces with specific characteristics.
- **Micro/Nanostructuring Techniques:** Various fabrication methods can be used to create micro- and nanostructures on surfaces, which contribute to super hydrophobicity. Examples include:
  - o **Photolithography:** Photolithography uses light-sensitive materials (photoresists) to pattern surfaces at the microscale or nanoscale. These patterns can be transferred onto the substrate to create controlled roughness.
  - o **Laser Ablation:** Laser ablation involves using a laser to remove or modify material on the surface, creating micro- or nanoscale features. This technique can

generate rough structures and surface textures that enhance super hydrophobic properties.

- **Nanosphere Lithography:** Nanosphere lithography utilizes self-assembled monolayers of closely packed nanospheres as a mask to create ordered nano scale patterns on the substrate. These patterns can be transferred into the substrate material to achieve superhydrophilicity.
- **Electrospinning:** Electrospinning involves using an electric field to draw a polymer solution into fine fibers. These fibers can be collected onto a substrate, creating a porous and rough surface structure suitable for super hydrophobic applications.
- **Chemical Modification:** Surface functionalization with hydrophobic molecules, such as alkylsilanes (e.g., octadecyl trichlorosilane, OTS), can be employed to reduce the surface energy and create super hydrophobicity. This technique involves depositing a self-assembled monolayer (SAM) of the hydrophobic molecules onto the substrate.

These are just a few examples of the techniques used to prepare super hydrophobic surfaces. Each technique has its advantages, and the choice depends on the specific material, substrate, and desired surface characteristics. Often, a combination of techniques is used to achieve optimal super hydrophobic properties.

### Engineering Applications of Super Hydrophobic Surfaces

Super hydrophobic surfaces have potential applications in the electronics, automobile, and aerospace industries, offering several benefits in these sectors. Here are some specific applications:

### Electronics Industry:

- **Waterproofing Electronics:** Super hydrophobic coatings can protect electronic components from water damage. By applying super hydrophobic coatings on circuit boards, connectors, and other sensitive electronic parts, water ingress can be minimized, improving the reliability and durability of electronic devices.
- **Moisture Resistance:** Electronic devices exposed to humid environments or moisture-prone conditions benefit from super hydrophobic coatings. These coatings prevent moisture from reaching and harming electronic components, reducing the risk of short circuits, corrosion, and malfunction.
- **Self-Cleaning Displays:** Super hydrophobic coatings applied to displays and touch screens repel water, oils, and fingerprints, making them easier to clean and maintain. This improves the visibility and functionality of electronic displays, especially in outdoor or light-touch application.

### Automobile Industry

- **Anti-Fogging Windows and Mirrors:** Super hydrophobic coatings can be used on automobile windows and mirrors to prevent fogging or condensation formation. The water-repellent property helps maintain clear visibility, enhancing driver safety and comfort in humid or cold weather conditions.
- **Self-Cleaning Surfaces:** Applying super hydrophobic coatings to the exterior surfaces of vehicles can facilitate self-cleaning by repelling water, dirt, and contaminants. This reduces the need for frequent washing and maintenance, keeping the vehicle cleaner and improving its appearance.
- **Fuel Efficiency:** Super hydrophobic coatings can reduce drag and frictional resistance on vehicle surfaces, leading to improved aerodynamics and fuel efficiency. By minimizing water adhesion, the coatings help reduce the accumulation of water droplets on the vehicle's exterior, decreasing drag and optimizing performance.

### Aerospace Industry:

- **Anti-Icing and Deicing:** Super hydrophobic coatings applied to aircraft surfaces can prevent ice formation or facilitate ice removal. This is particularly important for critical areas such as wings, engine components, and sensors, helping to ensure safe operations and reducing the risk of ice-related incidents.
- **Drag Reduction:** Super hydrophobic coatings on aircraft surfaces can minimize frictional drag during flight, leading to improved fuel efficiency and reduced emissions. The water-repellent property helps maintain a smooth airflow over the surface, optimizing aerodynamic performance.
- **Corrosion Resistance:** Super hydrophobic coatings can protect aerospace components from corrosion caused by exposure to moisture, rain, or harsh environments by repelling water and reducing surface contact with corrosive agents. These coatings help preserve the structural integrity and lifespan of aerospace equipment.

### 4.4.2 Self-Cleaning Surfaces

Self-cleaning surfaces are surfaces that are able to clean themselves without the need for manual cleaning. These surfaces are typically super hydrophobic and have a high contact angle with water, which causes water droplets to bead up and roll off the surface, carrying away any dirt or debris.

#### Principle of Self-Cleaning Surfaces

The principle of self-cleaning surfaces is based on two main mechanisms: the reduction of surface energy and the modification of surface texture. These mechanisms work together to minimize the adhesion of (1111 water, and oilier contaminants, enabling the self-cleaning effect. Here's a breakdown of the principle:

- **Low Surface Energy:** Self-cleaning surfaces often have low surface energy, which means they have a reduced affinity for liquid and solid particles. Materials with low surface energy repel water, oils, and other substances, preventing them from adhering

to the surface. This property is typically achieved through the application of hydrophobic or oleophobic coatings, such as fluoropolymers or other low-surface-energy materials.

- **Lotus Effect:** The Lotus Effect is a phenomenon observed in nature on the leaves of lotus plants. It is a classic example of self-cleaning surfaces. Lotus leaves have a unique micro/nanostructured surface covered with hydrophobic wax crystals. When water droplets come into contact with the leaf surface, they form near-perfect spheres and roll off, collecting dirt and contaminants along the way. This is due to the combination of the surface's low surface energy and the presence of micro/nanostructures, which reduce the contact area and enable easy droplet mobility.
- **Micro/Nanostructured Surfaces:** Surface texture plays a crucial role in self-cleaning surfaces. Microscopic or nanoscopic structures can be engineered or naturally occurring on a surface to create a roughness that limits the contact between the surface and contaminants. These structures can trap air pockets, causing liquids to form droplets with reduced contact area, minimizing adhesion. The trapped air can act as a lubricant, aiding in the easy removal of particles.
- **External Factors:** While the surface properties contribute to self-cleaning, external factors like water, wind, or light often play a role in activating the self-cleaning process. For example, the presence of water, either through rainfall or manual washing, can help remove loosely adhered particles from the surface. Sunlight or UV radiation can activate photocatalytic reactions on certain surfaces, breaking down organic matter and enhancing self-cleaning capabilities.

By combining low surface energy, micro/nanostructured surfaces, and external factors, self-cleaning surfaces minimize the adhesion and retention of contaminants making them easier to clean or enabling them to self-clean when exposed to appropriate conditions.

### Material and examples of Self-Cleaning Surfaces

Self-cleaning surfaces are designed to minimize the adhesion of dirt, dust, and other contaminants, making them easier to clean or allowing them to self-clean when exposed to external forces like water or sunlight. Here are some materials and examples of self-cleaning surfaces.

**Photocatalytic Coatings:** Photocatalyst material, such as titanium dioxide ( $\text{TiO}_2$ ), can be used as coating on surfaces to create self-cleaning properties. When exposed to ultraviolet (UV) light, the photocatalytic surfaces generate reactive oxygen species that break down organic matter, resulting in the decomposition of dirt and pollutants.

**Super hydrophobic Coatings:** Super hydrophobic surfaces exhibit extremely high-water repellency, which helps in the self-cleaning process. When water comes into contact with these surfaces, it forms spherical droplets that easily roll off, carrying away dirt and contaminants. Examples of super hydrophobic materials.

**Self-Cleaning Glass:** Self-cleaning glass incorporates a thin layer of titanium dioxide ( $\text{TiO}_2$ ) or other photocatalytic materials on the surface. When exposed to UV light, the photocatalytic reaction breaks down organic matter, while the hydrophilic nature of the surface allows water to spread and wash away the debris, resulting in a self-cleaning effect.

**Oleophobic Coatings:** Oleophobic surfaces repel oil and grease, making them resistant to stains and easier to clean. These Coating are typically made from fluorinated materials that have low surface energy, preventing oil or oily substances from adhering to the surface.

**Micro/Nanostructured Surfaces:** Surfaces with micro- or nanostructures can exhibit self-cleaning properties due to their ability to reduce contact area and enhance surface roughness. The Surfaces structures can trap air or create a lotus leaf-like effect, preventing the adhesion of dirt and facilitating self-cleaning when exposed to water or airflow.

**Self-Cleaning Fabrics:** Fabrics treated with hydrophobic or oleophobic coating can repel liquid, stains, and dirt making them easier to clean. These coating can be applied to textiles used in clothing, upholstery, or outdoor equipment, reducing the need for frequent washing and maintenance.

### Application of Self-cleaning surfaces and coatings

Self-cleaning surfaces have a wide range of applications in various industries. Here are some notable examples:

- **Architecture and Building Materials:** Self-cleaning surfaces find applications in architectural structures and building materials, such as self-cleaning glass for windows and facades. These surfaces repel dirt, dust, and pollutants, reducing the need for frequent cleaning and maintenance.
- **Solar Panels:** Self-cleaning coatings on solar panels prevent the accumulation of dust and dirt on the surface, ensuring optimal energy efficiency. By repelling contaminants, self-cleaning surfaces help maintain the transparency and effectiveness of solar panels.
- **Automotive Industry:** Self-cleaning surfaces can be applied to vehicle exteriors, including car windows and windshields. These surfaces repel water, oil, and dirt, improving visibility and reducing the need for frequent cleaning.
- **Electronics:** Self-cleaning coatings can be used on electronic displays, touchscreens, and optical lenses. These surfaces resist fingerprints, and smudges, ensuring clear visibility and enhancing device performance.
- **Textiles:** Self-cleaning coatings can be applied to fabrics used in outdoor clothing, upholstery, and carpets. These coatings repel liquids, stains, and dirt, making the textiles easier to clean and maintain.
- **Medical Equipment:** Self-cleaning surfaces can be utilized in medical equipment, such as hospital furniture, beds, and surfaces prone to contamination. These surfaces minimize the adhesion of microorganisms, reducing the risk of cross-contamination and improving hygiene.

- **Kitchen and Bathroom Surfaces:** Self-cleaning surfaces can be employed in kitchen countertops, sinks, and bathroom fixtures to repel water, oils, and stains. This helps keep the surfaces clean and reduces the effort required for cleaning and maintenance.
- **Outdoor Signage and Billboards:** Self-cleaning coatings on outdoor signage and billboards prevent the accumulation of dirt, grime, and pollutants. This helps maintain the visibility and effectiveness of advertisements, reducing the need for manual cleaning.
- **Air Conditioning and Ventilation Systems:** Self-cleaning coatings can be applied to air conditioning and ventilation system components, such as filters and ducts. These surfaces repel dust and particles, improving air quality and reducing the need for frequent cleaning or filter replacements.
- **Food and Beverage Industry:** Self-cleaning surfaces can be used in food processing equipment and containers to prevent the adhesion of food residues, oils, and contaminants. This enhances food safety and facilitates easier cleaning and sanitation.

The engineering applications of self-cleaning surfaces are vast and varied. The ability to repel dirt, dust, water, and oils offers advantages in terms of cleanliness, efficiency, and maintenance across numerous industries. By reducing the need for manual cleaning and improving the performance of various products and applications, self-cleaning surfaces have the potential to improve efficiency, reduce costs, and enhance safety across a range of industries.

### 4.5 Plants Burs and Velcro

Plant burs, such as those found on burdock, inspired the invention of Velcro, a popular hook-and-loop fastening system.

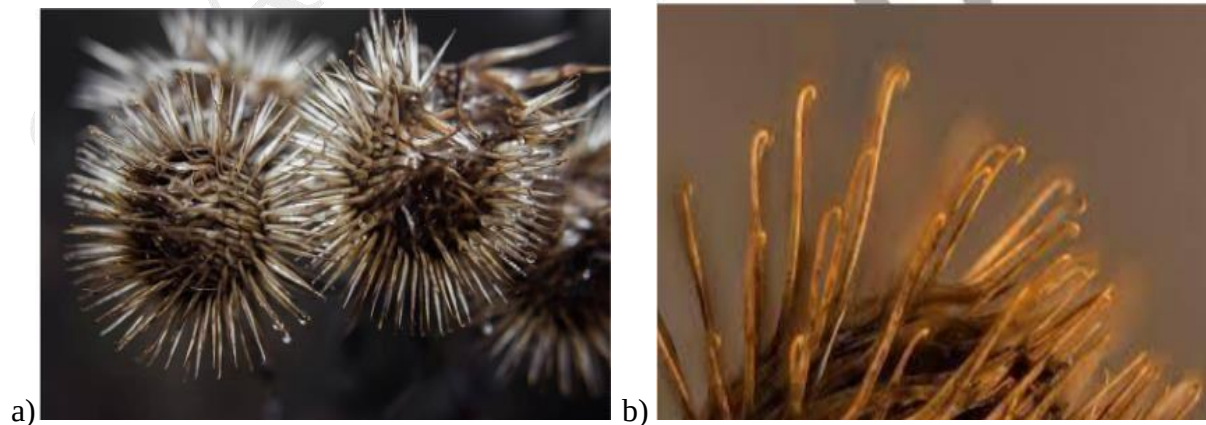


Fig: a) The Globular Flower heads of burdock, b) indicating the hook shape

The Burrs have small hooks that can latch onto clothing, fur, or feathers, allowing them to disperse their seeds over a wider area.

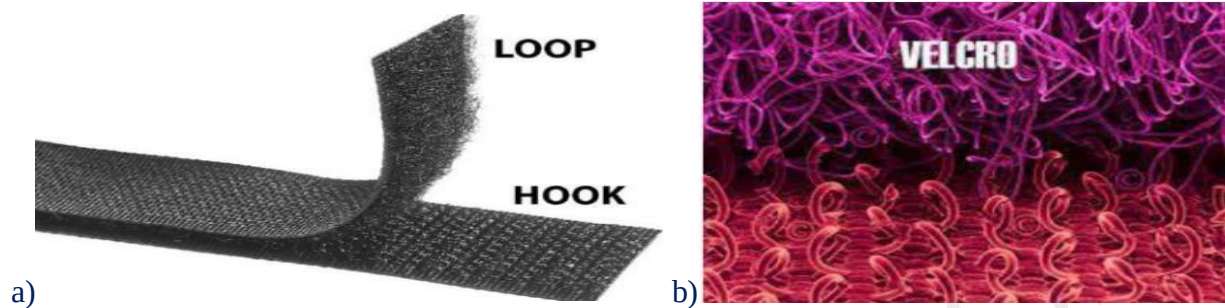


Fig: Image Showing a) hooks and loops normal view of Velcro, b) Microscopic view hooks and loops of Velcro.

Velcro was invented by Swiss engineer George De Mestral in 1941, after he became fascinated by the way buns clung to his clothes and his dog's fur during a walk.

He examined the buns under a microscope and found that they had small hooks that could latch onto loops in fabric.

De Mestral spent years experimenting with different materials before finally developing Velcro, which consists of two strips of nylon fabric, one with tiny hooks and the other with small loops. When pressed together, the hooks latch onto the loops, creating a strong bond that can be easily detached by pulling the two strips apart. Velcro has a wide range of applications, including in clothing, shoes, bags, and medical devices. It has become a popular alternative to traditional fasteners, such as buttons and zippers, due to its ease of use and versatility.

The name "Velcro" is actually a combination of the word's "velvet" and "crochet," as the fabric strips resemble velvet and are hooked together like crochet. Velcro has since become a popular alternative to traditional fasteners, such as buttons and zippers, due to its ease of use and versatility.

### Material Used in Velcro Technology

Velcro Technology uses two main materials: **nylon** and **Polyester**.

1. The nylon is extruded to create tiny hooks that are then cut and shaped into the familiar hooks shape. These hooks are designed to latch onto the loop side of the Velcro



Fig: The Hook of Velcro

The loop side of Velcro is made of polyester. Polyester is a Synthetic fabric that is strong and durable. The polyester is woven into a fabric that has many tiny loops. When the loops are pressed against the hook side of the Velcro, The Hooks latch onto the loops, Creating a secure attachment.



Fig: The Loop of Velcro.

In addition to nylon and polyester, the adhesive used to attach the Velcro to Surfaces can also vary. Some types of types of Velcro use a pressure-sensitive adhesive that can be easily removed without leaving a residue, while others use a stronger adhesive that creates a more permanent bond.

### Engineering Applications of Velcro Technology

**Clothing and footwear:**

Velcro is commonly used in clothing and footwear for closures and adjustable straps. It can be easily opened and closed, making it convenient for users with limited dexterity or mobility.

### **Medical devices:**

Velcro is used in medical devices such as braces, splints, and compression garments for its adjustable and secure fastening capabilities.

### **Aerospace equipment:**

Velcro is used in aerospace equipment, such as satellites any air spacecraft, to secure components in place and prevent them from vibrating or shifting during launch or flight.

### **Automotive industry:**

Velcro is used in the automotive industry for a range of applications such as securing carpets and headliners, and attaching door panels and seat cushions.

### **Packaging industry:**

Velcro is used in the packaging industry for reasonable closure on bags, pouches, and other types of packaging.

### **Sports equipment:**

Velcro is used in sports equipment's such as helmets and gloves, for its ability to provide a secure and adjustable.

## **4.6 Shark Skin and Friction Reducing Swim Suits**

The denticles on shark skin have evolved over millions of years to reduce drag and increase swimming efficiency. These structures disrupt the flow of water around the shark's bod, reducing turbulence and minimizing the formation of vortices. As a result, sharks can swim faster and with less effort compared to other fish.



**Fig: Indicating the denticles on shark skin.**

Denticles on shark skin are like tiny bumps or ridges. They disrupt the flow of water around the shark's body, making it smoother and reducing turbulence. This disruption reduces the resistance the shark experiences as it swims, allowing it to move faster and with less effort.

### **Turbulence in Water**

Turbulence is when a fluid, like water or air, becomes chaotic and unpredictable. Instead of flowing smoothly, it swirls and forms irregular patterns. This turbulence creates resistance or drag, which makes it harder for things to move through the fluid. In swimming, reducing turbulence is important because it helps to minimize resistance, swimmers to move more easily and efficiently through the water.

### **Reducing Drag**

When a shark swims through the water, the water normally flows smoothly over its body. However, the denticles on the shark's skin disrupt this smooth flow. They create small disturbances in the water, which helps to break up turbulent currents that can slow the shark down. By reducing turbulence, the denticles make the flow of water around the shark's body smoother. This smoother flow reduces the resistance or drags the shark experiences as it moves through the water, allowing it to swim more efficiently.

### Frictionless Swim Suits

Shark skin has inspired the development of friction-reducing swim suits, which are designed to improve the performance of swimmers by reducing drag in the water.

Friction-reducing swim suit use a similar structure to that of shark skin to reduce drag and improve swimmer performance these suits are made from high-tech materials that mimic the properties of shark skin, such as the shape and size of the denticles.

### Materials Used

The materials used to friction-reducing swim suits inspired by shark skin include:

- **Polyurethane** A type of per that is commonly used in the production of swim suits, as it is durable and can be molded to a variety of shapes.
- **Lycra/ Spandex:** Lycra and spandex are made from the same synthetic fiber, which is technically called elastane. Elastane fibers are typically composed of a polymer called polyurethane which is then blended with other fibers like nylon, polyester, or cotton) that is known for its stretch and flexibility.
- **High-tech fabrics:** A range of high-tech fabrics have been developed specifically for use in swim suits. These fabrics are designed to be lightweight, water-repellent, and hydrodynamic, and often incorporate materials such as silicone or Teflon to reduce drag.

### Examples

- **Speedo Fastskin:** This swim suit was designed based on the structure of shark skin and is made from a high-tech fabric that incorporates a range of materials to reduce drag and turbulence in the water.
- **Arena Powerskin Carbon Ultra:** Another example of a friction-reducing swim suit, the Arena Powerskin Carbon Ultra is made from a combination of polyurethane and high-tech fabrics to provide a hydrodynamic and form-fitting design.
- **TYR Venzo:** The TYR Venzo is a friction-reducing swim suit that incorporates a unique surface structure inspired by shark skin, as well as other advanced materials to improve swimmer performance.

#### **4.7 Kingfisher Beak and Bullet Train**



Figure: Indicating the shape similarities of kingfisher beak and design of the front of the bullet train

The kingfisher beak is an excellent example of nature's design for efficient diving and fishing. Its unique shape and structure enable the kingfisher to minimize the impact of water resistance and achieve a successful dive.

#### **The Physics behind the Kingfisher Beak**

##### **Streamlining:**

The beak of a kingfisher is long, slender, and sharply pointed, which helps reduce drag or air resistance as the bird dives into the water. The streamlined shape allows the kingfisher to smoothly cut through the air and minimize the energy required for the dive.

##### **Surface Tension:**

When the kingfisher hits the water, it encounters the resistance caused by surface tension. Surface tension is the cohesive force between water molecules that creates a "skin" on the water's surface. The sharp beak of the kingfisher helps to pierce through the water's surface, breaking the surface tension and reducing the force required to enter the water.

##### **Minimizing Splash:**

As the kingfisher dives, it needs to enter the water with minimal disturbance to avoid scaring away the fish it intends to catch. The shape of the beak helps to

reduce the splash generated upon entry. The beak's narrow and pointed design helps create a smooth entry by minimizing the disturbance of the water surface, allowing the kingfisher to enter silently and effectively.



Fig: Image of a Shinkansen bullet train of Japan.

The use of the kingfisher beak as a design inspiration for the front of the bullet train is an example of how nature-inspired engineering can lead to innovative solutions that improve the performance and efficiency of a Shinkansen bullet train of Japan is the best example which used the biomimicry of kingfisher's beak.

### **Aerodynamic Design:**

The front of the Shinkansen is meticulously shaped to reduce air resistance and improve aerodynamic performance. The streamlined design minimizes drag as the train travels at high speeds, allowing it to maintain stability and efficiency. The smooth, tapered shape reduces the pressure difference between the front and rear of the train, reducing noise and vibration.

### **Pressure Wave Reduction:**

When a high-speed train moves through a tunnel, it creates pressure waves that can cause noise and discomfort for passengers. The nose of the Shinkansen is designed to reduce these pressure waves by effectively managing airflow and minimizing the compression and expansion of air as the train enters and exits tunnels. This reduces the noise level and enhances passenger comfort.

### 4.8 Human Blood Substitutes Introduction

Human blood substitutes are synthetic products that are designed to act as a replacement for blood in the human body.

#### **Basic Requirement for Human Blood Substitutes**

##### **Effective Oxygen Transport:**

Human blood substitutes must be capable of efficiently carrying and delivering oxygen to the body's tissues. This is a fundamental function of natural blood that any substitute should be able to replicate or improve upon.

##### **Safety and Compatibility:**

Blood substitutes should be safe for use in the human body and well-tolerated by the recipient. They should not cause significant adverse reactions, toxicity, or immune responses. Additionally, they should not interfere with normal blood clotting or other essential physiological processes.

##### **Storage and Transport:**

Human blood substitutes should be stable and capable of being stored and transported easily. This is particularly important in emergency situations or areas where access to blood products may be limited. The ability to store and transport substitutes effectively ensures their availability when needed.

##### **Cost-Effectiveness and Scalability:**

Blood substitutes should be cost-effective and sealable for widespread use in medical settings. They should be affordable and feasible to produce in large quantities, meeting the potential demand for blood products.

#### **Types of HBS**

There are two types of human blood substitutes - hemoglobin-based oxygen carriers (HBOCs) and perfluorocarbons (PFC's)

HBOCs are based on the hemoglobin molecule, which is the protein in red blood cells that carries oxygen to the body's tissues. Hemoglobin is extracted from human or animal blood and then modified to create a stable, synthetic version. When introduced into the body, HBOCs can help to increase amount of oxygen available to the tissues, which can be important in situation body is unable to produce or transport enough red blood cells.

PFC's are synthetic molecules that are similar in structure to the hemoglobin molecule. However, unlike HBOC's they do not require modification from natural sources. PFCs are able to dissolve oxygen and transport it throughout the body, similar to the way that red blood cells work.

### **Advantages of hemoglobin-based oxygen carriers**

#### **Increased oxygen-carrying capacity:**

HBOCs can potentially carry more oxygen per unit volume than whole blood. This can be advantageous in situations where there is a need for rapid oxygen delivery or when there is limited availability of blood for transfusion.

#### **Universal compatibility:**

Unlike blood transfusions, which require blood typing and cross-matching to ensure compatibility, HBOCs can potentially be universally compatible with any blood type. This can be particularly useful in emergency situations or in areas where blood matching facilities are limited.

#### **Longer shelf life:**

HBOCs have the potential for longer storage and shelf life compared to donated blood, which has a limited lifespan. This can improve availability of oxygen-carrying substitutes in critical situations and reduce the need for frequent blood donations.

**Reduced risk of infections:** Blood transfusions carry a small risk of transmitting infection, such as viruses or bacteria, from the donor to the recipient since HBOC's are synthetic and

do not rely on human donors, the risk of infections associated with transaction can be significantly reduced.

### **Availability in Remote or Challenging settings:**

In remote or underdeveloped area where access to safe blood transaction may be limited HBO's can potentially provide a viable alternative for oxygen delivery. This can be particularly beneficial in military setting disaster relief efforts, or during transport of patients where immediate access to blood is not feasible.

### **Limitations/Risks of using HBOCs**

#### **Limited oxygen release:**

One of the challenges with HBOCs is ensuring efficient oxygen release to the tissues. The oxygen dissociation curve of HBOCs may differ from that of natural red blood cells, potentially leading to inadequate oxygen delivery to tissues in certain conditions.

#### **Short half-life:**

HBOCs tend to have a shorter half-life in the body compared to natural red blood cells. This means that the HBOCs may be rapidly cleared from circulation, reducing their effectiveness and requiring more frequent doses or infusions.

#### **Nitric oxide scavenging:**

HBOCs have a tendency to scavenge nitric oxide, a molecule important for regulating blood vessel dilation and maintaining normal blood flow. Excessive nitric oxide scavenging by HBOCs can lead to vasoconstriction, impairing blood flow to vital organs and potentially causing adverse cardiovascular effects.

#### **Renal toxicity:**

Some HBOCs have shown a potential for renal toxicity, causing damage to the kidneys. This can be a significant concern as the kidneys play a crucial role in filtering and excreting waste products from the body.

### **Immunogenicity and adverse reactions:**

HBOCs can trigger immune responses in the body, potentially leading to allergic reactions or other adverse events. Immunogenicity can vary between different HBOC products and individuals, and careful monitoring is necessary to identify and manage any potential adverse reactions.

### **Regulatory challenges:**

HBOCs are subject to rigorous regulatory scrutiny due to their potential risks and complex nature. Obtaining regulatory approval for HBOCs can be a lengthy and costly process, and several HBOC products have faced setbacks in their development due to safety concerns.

### **Interference with diagnostic tests:**

HBOCs can interfere with certain laboratory tests, such as those measuring bilirubin or liver enzymes. This interference can complicate the interpretation of results and potentially lead to diagnostic errors.

### **Examples of HBOCs**

There are several examples of hemoglobin-based oxygen carriers (HBOCs) that have been developed or are currently in development. Here are a few examples:

- **Hemopure:** Hemopure is an HBOC that is made from bovine hemoglobin. It has been approved for use in South Africa, Russia, and some other countries.
- **Oxyglobin:** Oxyglobin is another HBOC that is made from bovine hemoglobin. It is approved for veterinary use in the United States and has been used to treat anemia in dogs.
- **Hemospan:** Hemospan is an HBOC that is being developed by Sangart Inc. It is currently in clinical trials and has shown promise in increasing oxygen delivery to tissues.

- **MP4OX:** MP4OX is an HBOC that is being developed by Baxter Healthcare. It is designed to increase oxygen delivery to tissues and also to scavenge harmful free radicals in the bloodstream.
- **Hemolink:** Hemolink is an HBOC that is being developed by Hemosol Inc. It is designed to be used in trauma and surgical settings and has shown promise in improving oxygen delivery to tissues.

(Note: Many countries have not yet given regulatory approval for clinical usage of HBOCs)

### 4.8.2 Perfluorocarbons (PFCs)

Perfluorocarbons (PFCs) are a type of human blood substitute that are designed to deliver oxygen to the body's tissues. Unlike hemoglobin-based oxygen carriers (HBOCs), which are based on natural proteins, PFCs are synthetic chemicals that are similar in structure to some types of industrial solvents.

#### Advantages of PFC's

##### High Oxygen-carrying capacity:

PFC's have the ability to dissolve a significant amount of oxygen much higher than that of blood. This allows for efficient oxygen delivery to tissues, even in low-oxygen environments.

##### Improved oxygen solubility:

PFC's Exhibit a high solubility for oxygen, meaning that oxygen molecules can readily dissolve in PFC solutions. This enables PFC's to transport and deliver oxygen more effectively than other alternatives.

##### Stability and long shelf life:

PFC's are chemically stable and have a long shelf life, making them suitable for storage and use in emergency situation where the availability of fresh blood or other oxygen more effectively than other alternatives.

### **No blood typing or cross-matching required:**

Unlike blood transactions which require compatibility testing and matching of blood types, PFC's are not dependent on blood typing this makes them potentially universal oxygen carries, suitable for use in individuals of any blood type.

### **Reduced risk of infection transmission:**

PFC's are synthetic substances, eliminating the risk of transmitting infectious disease associated with blood transactions. This advantage can be particularly significant in situations where the availability of safe blood products is limited or in area with a high prevalence of blood-borne infections.

### **Compatibility with diagnostic tests:**

PFC's do not interface with laboratory diagnostic tests, allowing for accurate interpretation of test results without potential complications from the presence of PFC's.

### **Limitations of PFCs**

#### **Limited oxygen offloading:**

While PFCs have a high capacity to carry and dissolve oxygen, they tend to have a reduced ability to release oxygen to tissues compared to red blood cells. This can result in inefficient oxygen delivery, especially in situations where oxygen demand is high or oxygen tension in tissues is low.

#### **Need for specialized administration methods:**

PFCs typically require specialized administration techniques, such as emulsification or encapsulation, to enhance their stability and improve their oxygen-carrying capacity. These techniques can add complexity and cost to the administration process.

### **Short half-life:**

PFCs have a relatively short half-life in the body, leading to the need for frequent administration to maintain adequate oxygen-carrying capacity. This can be impractical in certain clinical scenarios or situations where prolonged oxygen delivery is required.

### **Clearance and elimination:**

PFCs are primarily eliminated from the body through the kinetics can vary among individuals. This can impact their effectiveness and clearance rates potentially limiting their duration of action.

### **Side effects and toxicity:**

PFCs have the potential for side effects and toxicity particularly if used in excessive amounts or for prolonged periods. Adverse effects can include respiratory distress, immune reactions, and potential organ toxicity. The safety profile of PFCs needs to be thoroughly studied and monitored.

### **Regulatory considerations:**

PFCs are subject to regulatory approval and scrutiny, like other medical products. Obtaining regulatory approval for PFC-based products can involve extensive testing and evaluation to ensure their safety and efficiency.

### **Examples of PFCs**

**Perftoran:** Perftoran is a PFC that was developed in Russia and is used in several countries, including Russia, Ukraine and China it has been used in the treatment of a variety of conditions, including trauma, heart attack, and stroke.

**Oxycyte:** Oxycyte is a PFC that is being developed by Oxygen Biotherapeutics. It is currently in clinical trials and has shown promise in increasing oxygen delivery to tissue in patients with traumatic brain injury.

**Oxycyte PFC Emulsion:** This is another PFC based blood substitute being developed by oxygen biotherapeutics. It is designed to be used as an oxygen carrier during surgery and other medical procedures.

**Hemopure-PFC:** Hemopure-PFC is a hybrid blood substitute that combines a PFC with a hemoglobin-based oxygen carrier. It is being Developed by HbO2 therapeutics and has shown promise in increasing oxygen delivery to tissues in preclinical studies.

It's important to note that while these technologies show promise they are still in developed and further studies are needed to evaluate their safety and effectiveness.

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